Run report: item1D-mult-run-20261018T2106Z

Descriptive record of one recorded run of examples/decomposition_coverage.rs, mode mult (Amendment D) under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G and Addendum H, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
commit 44a81248a6aaf64ac4d5373e0c97867675887e2f
cmd: dc research/weil_descent_angles_20261018/results/item1D-mult-run-20261018T2106Z mult
rustc 1.97.0 (2d8144b78 2026-07-07)
Sun Oct 18 21:06:02 UTC 2026
post-run (d2edb2ab8): the tool was rustfmt-formatted (format only; unit tests pass); outputs unaffected.
```

checks.txt

```
M1 identity x(P+Q)+x(P-Q) = x1x2/(x1+x2)^2 violations: 0
M2 solution cosets not a power of 2: 0
M3 cross-check distinct(subfield) != distinct from cosets: 0 curves
M3 dispersion char 0 null: mean 0.9923 sd 0.0213 over 15 curves
M3 dispersion char 0 subfield: mean 0.9725 sd 0.0320 over 15 curves
M3 dispersion char 2 null: mean 0.9576 sd 0.0955 over 21 curves
M3 dispersion char 2 subfield: mean 1.0007 sd 0.0301 over 21 curves
```

binary_cosets.csv (first 12 of 21 rows)

m	curve	order	m1_pairs	m1_violations	x3_total	solvable	nonpow2_cosets	kdim0	kdim1	kdim2	kdim3	kdim4plus	valid0	valid1	valid2	valid3	valid4	valid5	valid6	valid7	valid8plus	distinct_subfield	distinct_from_cosets
4	0	240	28	0	256	120	0	0	240	0	0	16	79	26	15	0	0	0	0	0	0	83	83
4	1	234	36	0	256	120	0	0	240	0	0	16	70	28	22	0	0	0	0	0	0	101	101
4	2	240	28	0	256	120	0	0	240	0	0	16	79	26	15	0	0	0	0	0	0	83	83
5	0	990	136	0	1024	496	0	0	992	0	0	32	291	138	67	0	0	0	0	0	0	411	411
5	1	990	136	0	1024	496	0	0	992	0	0	32	300	120	76	0	0	0	0	0	0	393	393
5	2	1034	136	0	1024	496	0	0	992	0	0	32	286	148	62	0	0	0	0	0	0	421	421
6	0	3982	528	0	4096	2016	0	0	4032	0	0	64	1235	506	275	0	0	0	0	0	0	1563	1563
6	1	4034	528	0	4096	2016	0	0	4032	0	0	64	1245	486	285	0	0	0	0	0	0	1543	1543
6	2	4096	496	0	4096	2016	0	0	4032	0	0	64	1269	502	245	0	0	0	0	0	0	1495	1495
7	0	16492	2016	0	16384	8128	0	0	16256	0	0	128	5075	2074	979	0	0	0	0	0	0	6107	6107
7	1	16230	2080	0	16384	8128	0	0	16256	0	0	128	5016	2064	1048	0	0	0	0	0	0	6225	6225
7	2	16408	2016	0	16384	8128	0	0	16256	0	0	128	5103	2018	1007	0	0	0	0	0	0	6051	6051

mult.csv (first 12 of 72 rows)

char	q	curve	order	arm	F	lambda	poisson	pi	dispersion	h0	h1	h2	h3	h4	h5	h6	h7	h8plus
2	16	0	240	subfield	15	0.500000	0.393469	0.345833	1.06710	157	52	30	0	0	0	0	0	0
2	16	0	240	null	10	0.229167	0.204804	0.212500	0.79079	189	50	0	0	0	0	0	0	0
2	16	1	234	subfield	17	0.653846	0.479958	0.431624	0.99309	133	56	44	0	0	0	0	0	0
2	16	1	234	null	20	0.897436	0.592387	0.636752	0.82163	85	110	26	10	2	0	0	0	0
2	16	2	240	subfield	15	0.500000	0.393469	0.345833	1.06710	157	52	30	0	0	0	0	0	0
2	16	2	240	null	13	0.379167	0.315568	0.333333	0.76758	160	74	5	0	0	0	0	0	0
2	32	0	990	subfield	33	0.566667	0.432586	0.415152	0.94260	579	276	134	0	0	0	0	0	0

2	32	0	990	null	36	0.672727	0.489685	0.510101	0.83862	485	376	112	16	0	0	0	0	0
2	32	1	990	subfield	33	0.566667	0.432586	0.396970	1.00877	597	240	152	0	0	0	0	0	0
2	32	1	990	null	26	0.354545	0.298508	0.308081	0.87126	685	272	30	2	0	0	0	0	0
2	32	2	1034	subfield	33	0.542553	0.418738	0.407157	0.92926	613	296	124	0	0	0	0	0	0
2	32	2	1034	null	22	0.244681	0.217046	0.192456	1.16243	835	158	36	4	0	0	0	0	0

Graph

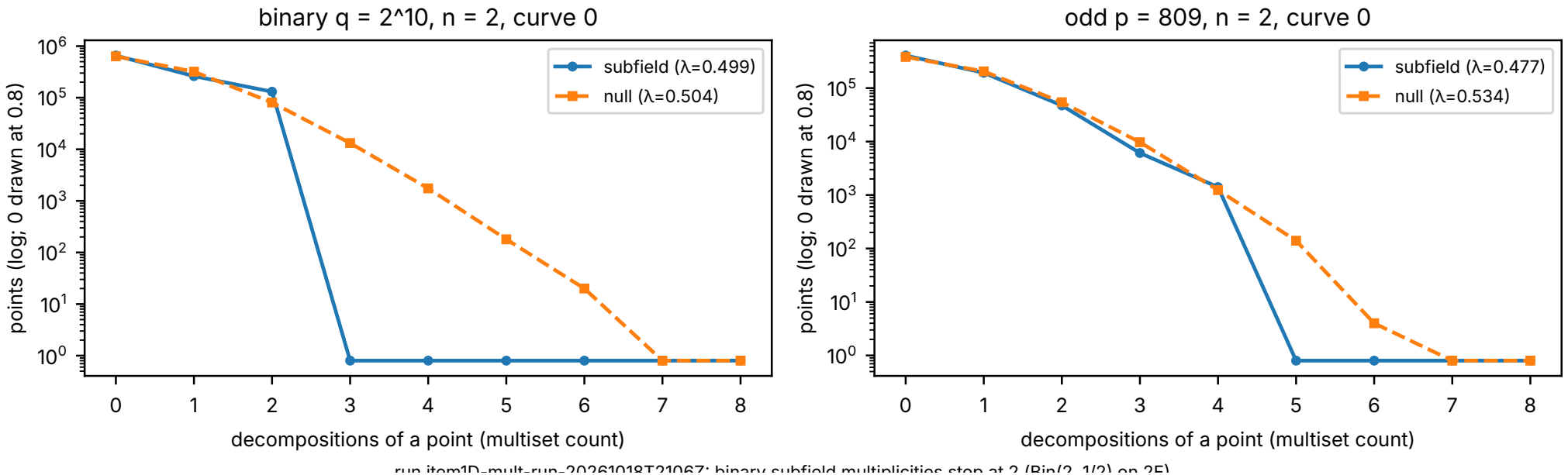


Figure 1: Amendment D: point multiplicity histograms, subfield vs null; exact.